Distributions and sampling: Gaussian, central limit theorem, A/B tests and temperature

Why can the same question get a different answer from a chat model, and how do we test whether a new AI feature is better?

Mathematics for AI and Machine Learning | Session 14 of 18 | Probability and statistics
KLAi Professional Learning | Jotham Martin Wambi

Today in one view: what you will leave with

By the end of today you can



Describe the Bernoulli, binomial and Gaussian distributions, and the multivariate Gaussian as a picture

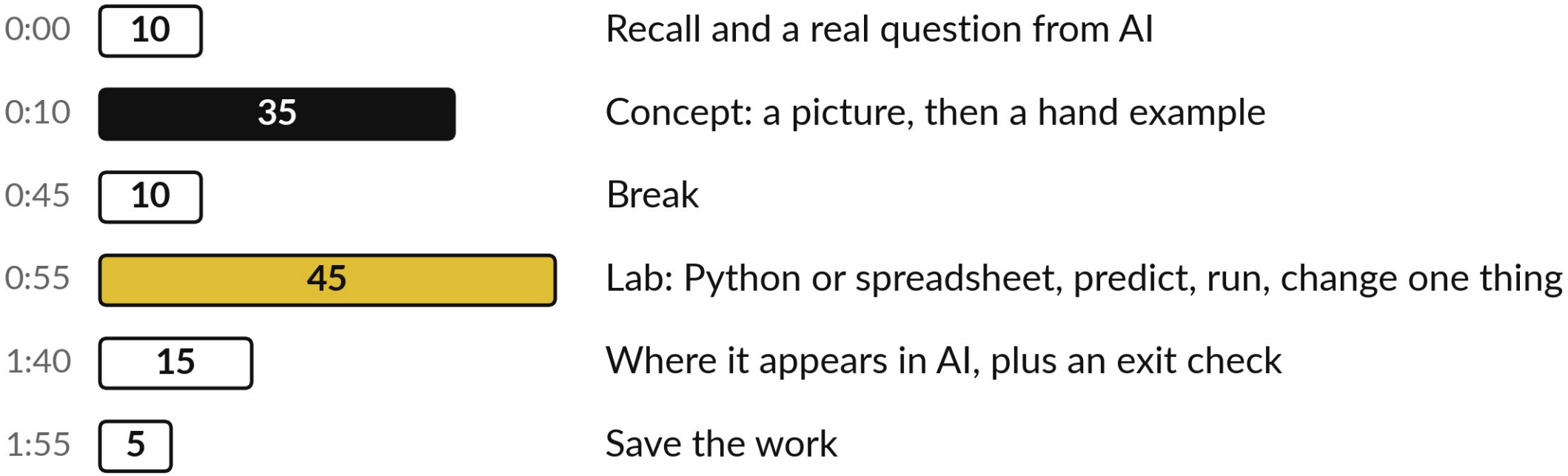


Explain the law of large numbers, the central limit theorem, standard error and a confidence interval, and run a two-proportion A/B test



Calculate softmax with temperature and a top-p cut on toy logits, and sample the next word from the result

Your two hours: distributions, tests, then sampling words



Pictures and hand calculations first. Then the lab on sampling, averages and temperature.

Three questions from last time

1. The probabilities of all possible next words must add up to what number, and which rule says so?

2. What is the expectation of a fair die, and what does it mean?

3. What is the difference between covariance and correlation?



Answer aloud first. Then we check together.

Answers to the three questions from Session 13

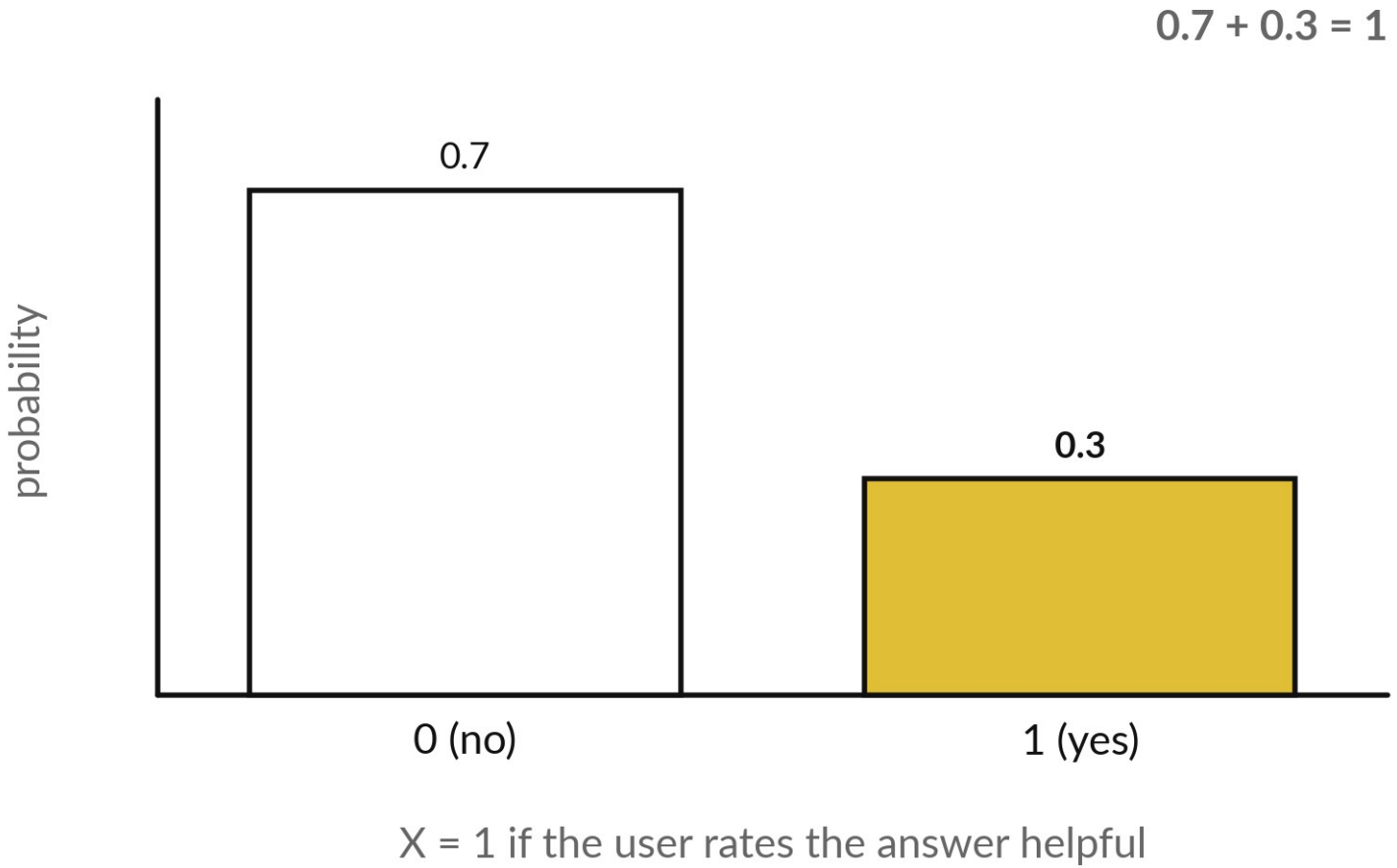
Question	Short answer
The probabilities of all possible next words must add up to what number, and which rule says so?	To 1. The total rule: all outcomes together have probability 1.
What is the expectation of a fair die, and what does it mean?	3.5. It is the probability-weighted average over many rolls. No single roll shows it.
What is the difference between covariance and correlation?	Covariance has units. Correlation divides by both standard deviations, has no units and lies between -1 and +1.

If an answer surprised you, look back at Session 13 before the lab.

Words we use today

	Word	What it means
	**Gaussian (normal) distribution**	The bell curve, set by its mean mu and standard deviation sigma.
	**Central limit theorem**	Averages of many independent values are close to Gaussian, whatever the original shape.
	**Standard error and confidence interval**	The standard deviation of a sample mean, and a range built from it that usually captures the true value.
	**Hypothesis test and p-value**	A check of whether a gap could be luck. The p-value is the probability of seeing a gap at least this large, in either direction, if there were truly no difference.
	**Temperature and top-p (nucleus) sampling**	Settings that reshape and trim the next-word probabilities before one word is drawn.

A Bernoulli random variable is one yes or no event with probability p



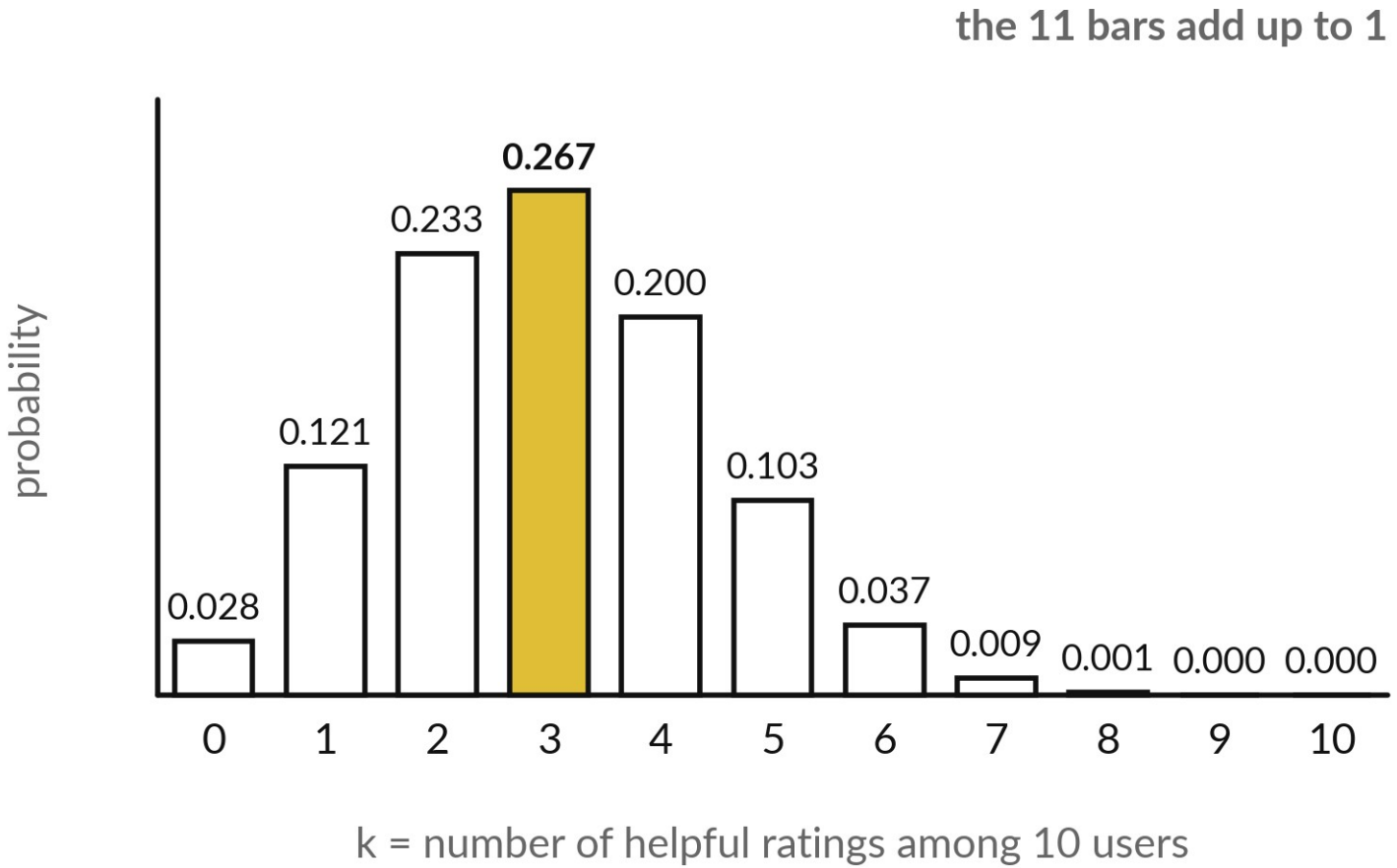
Illustrative: one user rates an answer helpful with probability p = 0.3.

Bernoulli(p): X is 1 with probability p and 0 with probability 1 - p.

E(X) = p = 0.3.

Var(X) = p(1 - p) = 0.21.

A binomial random variable counts the successes in n independent Bernoulli trials



Ten users, each rates helpful with p = 0.3 (illustrative). X counts the helpful ratings.

P(X = 3) = 0.2668, the gold bar.

Mean n p = 3. Standard deviation = 1.449.

Binomial(n, p) has a mass function with n + 1 bars.

The binomial formula counts the ways and multiplies the probabilities

Binomial pmf

$$P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}$$

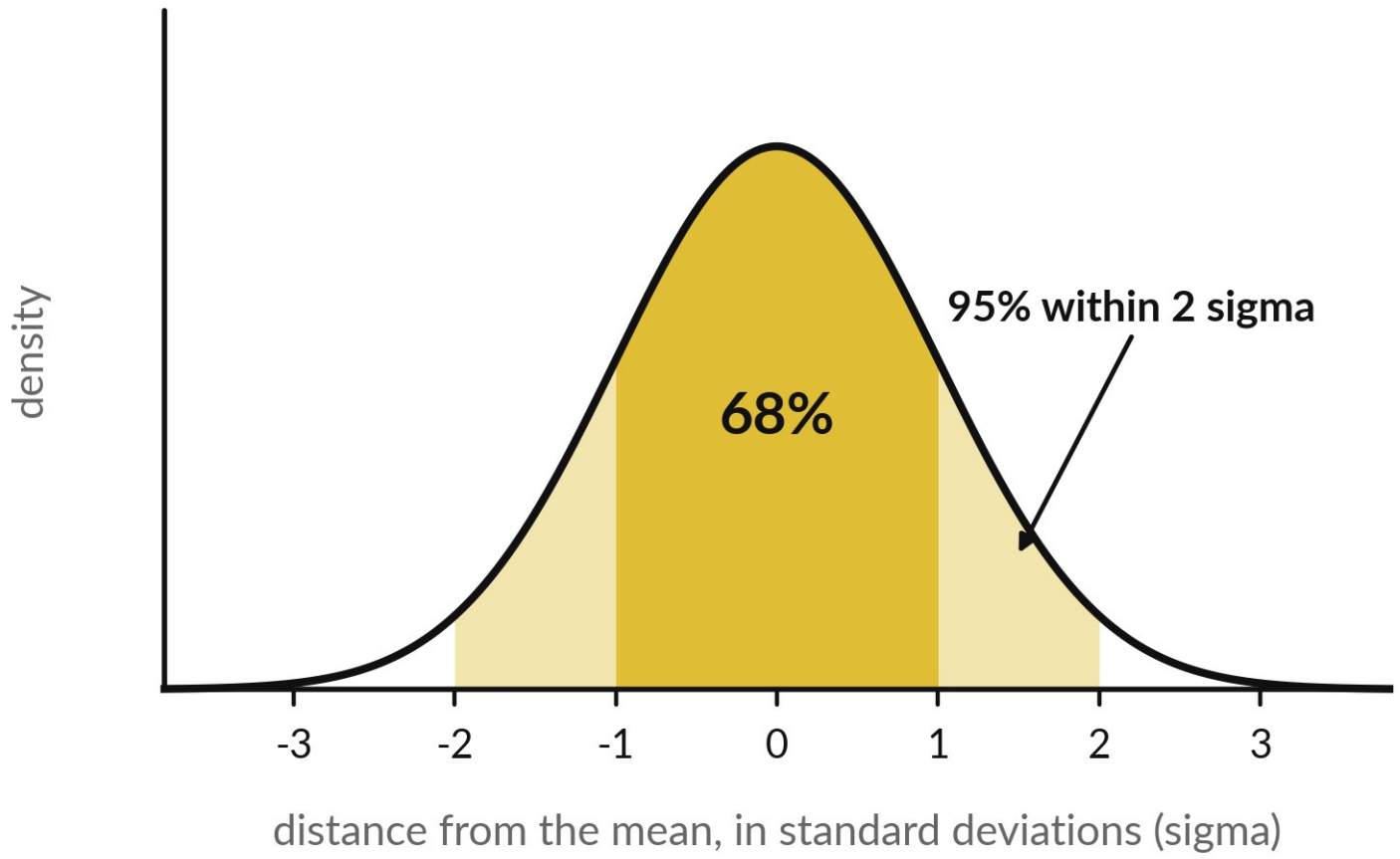
Mean and variance

$$E(X) = np, \qquad \mathrm{Var}(X) = np(1-p)$$

Symbol	What it means	In our data
n choose k	The number of ways to place k successes among n trials: n! / (k! (n-k)!).	10 choose 3 = 120
p to the k	Probability that k trials all succeed.	0.3 cubed = 0.027
(1-p) to the n-k	Probability that the other n - k trials all fail.	0.7 to the 7 = 0.08235
P(X = 3)	Multiply the three parts.	120 × 0.027 × 0.08235 = 0.2668

Spreadsheet: BINOM.DIST(3, 10, 0.3, FALSE) gives the same number. TRUE gives P(X at most 3).

The Gaussian (normal) distribution is the bell curve set by its mean and standard deviation



Gaussian (normal) distribution: symmetric, centred on the mean mu, with width set by sigma.

About 68% of values lie within 1 sigma of mu, and about 95% within 2 sigma.

Averages and sums of many small effects look Gaussian. The central limit theorem says why.

The Gaussian density and the z-score

Gaussian pdf

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

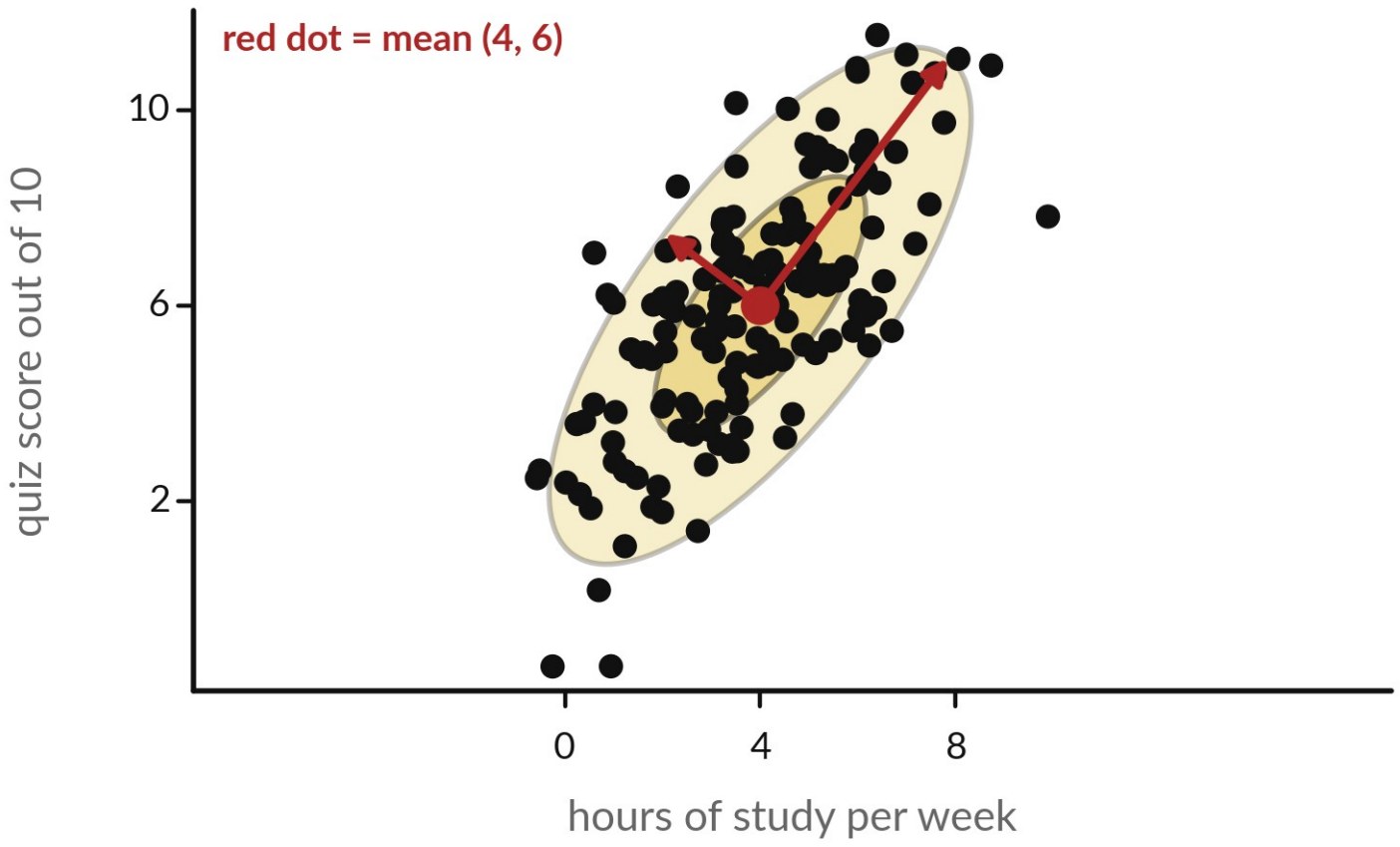
z-score

$$z = \frac{x - \mu}{\sigma}$$

Symbol	What it means	In our data
mu, sigma	Mean and standard deviation. Quiz scores (illustrative).	mu = 60, sigma = 10
z	How many standard deviations x lies above the mean.	x = 75 gives z = 1.5
Areas	P(X < 75) is the area left of z = 1.5. P(50 < X < 70) is the area between z = -1 and 1.	0.9332 and 0.6827

No integrating by hand: NORM.DIST in a sheet or math.erf in Python gives the areas.

A multivariate Gaussian is a cloud with a centre and a tilted ellipse



Two columns at once: study hours and score from Session 13 (illustrative). 150 simulated points.

Centre: the means, 4 and 6.

Ellipses: 1 and 2 standard deviations.

Red arrow = eigenvector of the covariance matrix, along an ellipse axis. Its eigenvalue is the variance along it.

The covariance matrix puts every variance and covariance in one table

	A	B	C
		hours	**score**
1	hours	4.667	4.167
2	score	4.167	7.000

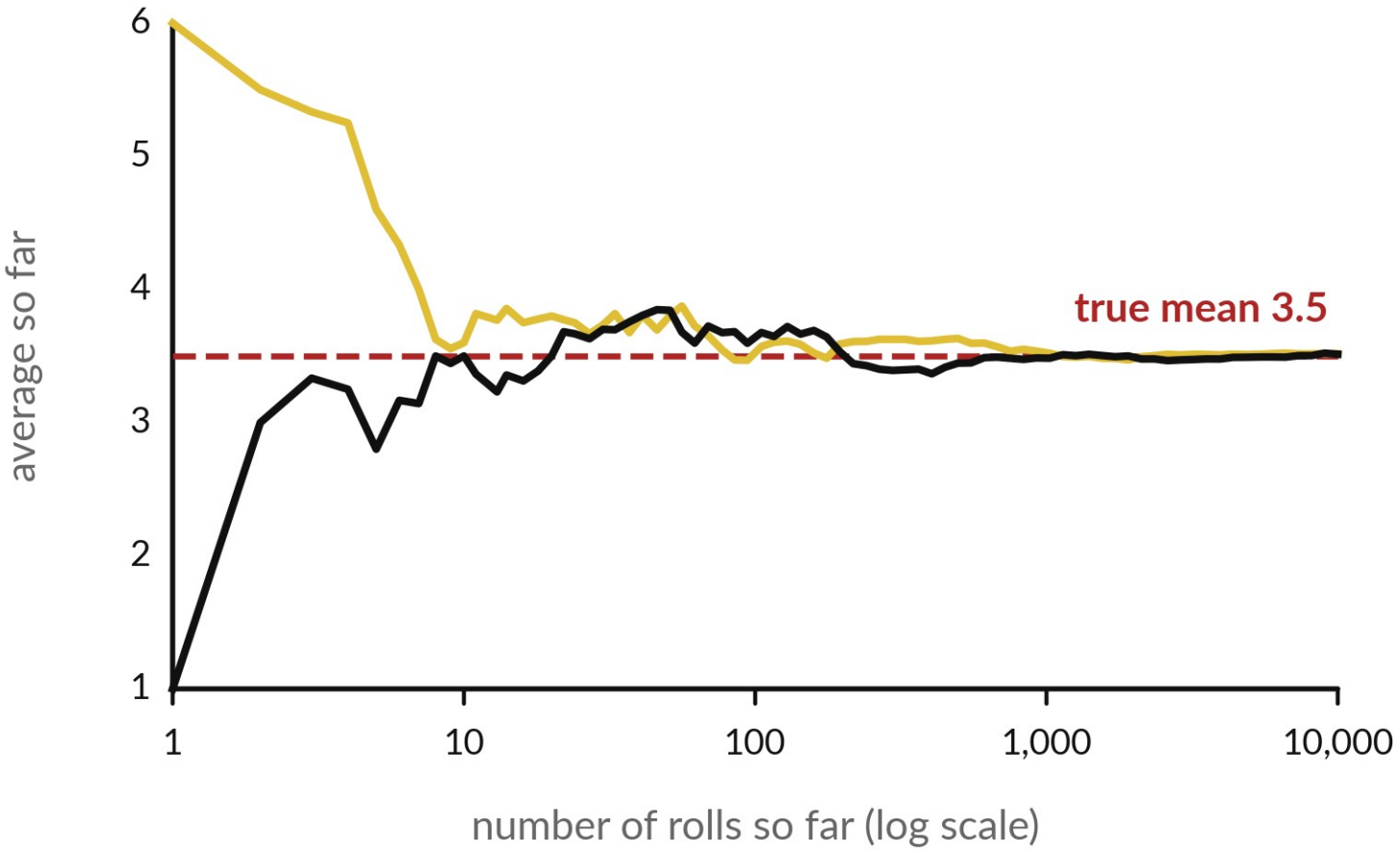
The covariance matrix of the two columns from Session 13 (this course divides by n).

Diagonal: the variances 4.667 and 7.

Off the diagonal: the covariance 4.167, twice.

Its eigenvalues are 10.160 and 1.506: the ellipse is long in one direction and thin in the other.

The law of large numbers: the more repeats, the closer the average gets to the true mean



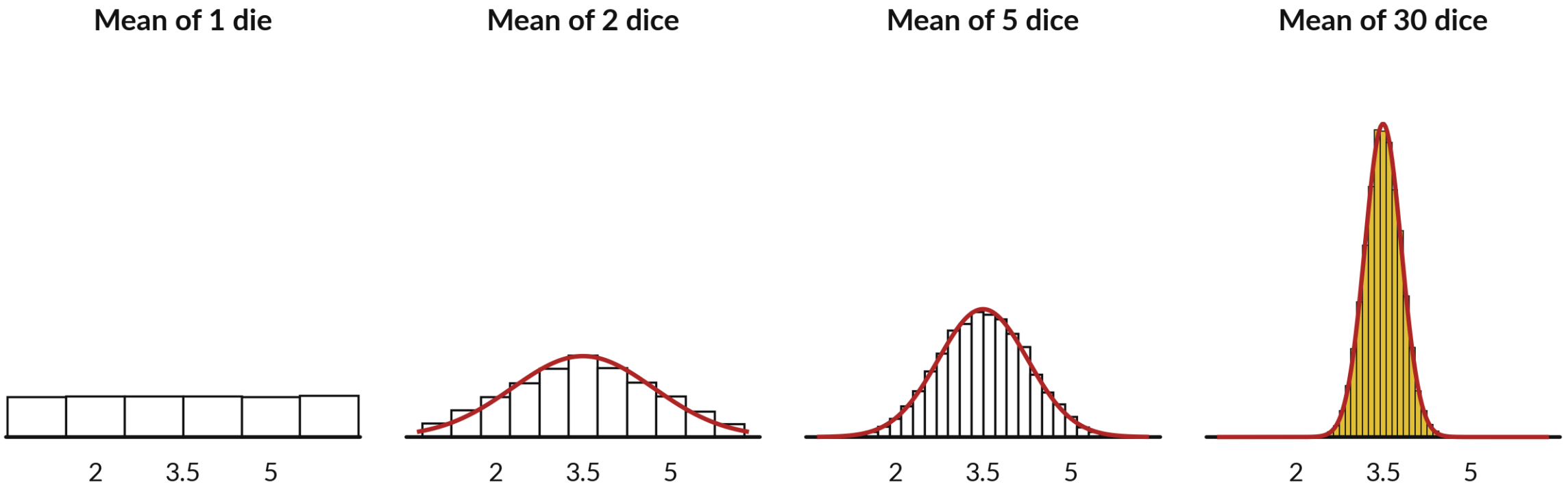
Two runs of rolling a die 10,000 times (simulation). The line is the running average.

After 1 roll the average is one of 1 to 6, often far from 3.5.

After 100 rolls it is near 3.5.

After 10,000 rolls it is usually within about 0.03 of 3.5.

The central limit theorem: averages of many rolls look Gaussian, though a single die is flat



Each panel: 20,000 averages of n dice. n = 1 is flat, not Gaussian. At n = 30 the red curve fits.

Standard error: how much a sample mean wanders

Central limit theorem

$$\bar{X} \approx \text{Gaussian}\left(\mu,\ \frac{\sigma^2}{n}\right)$$

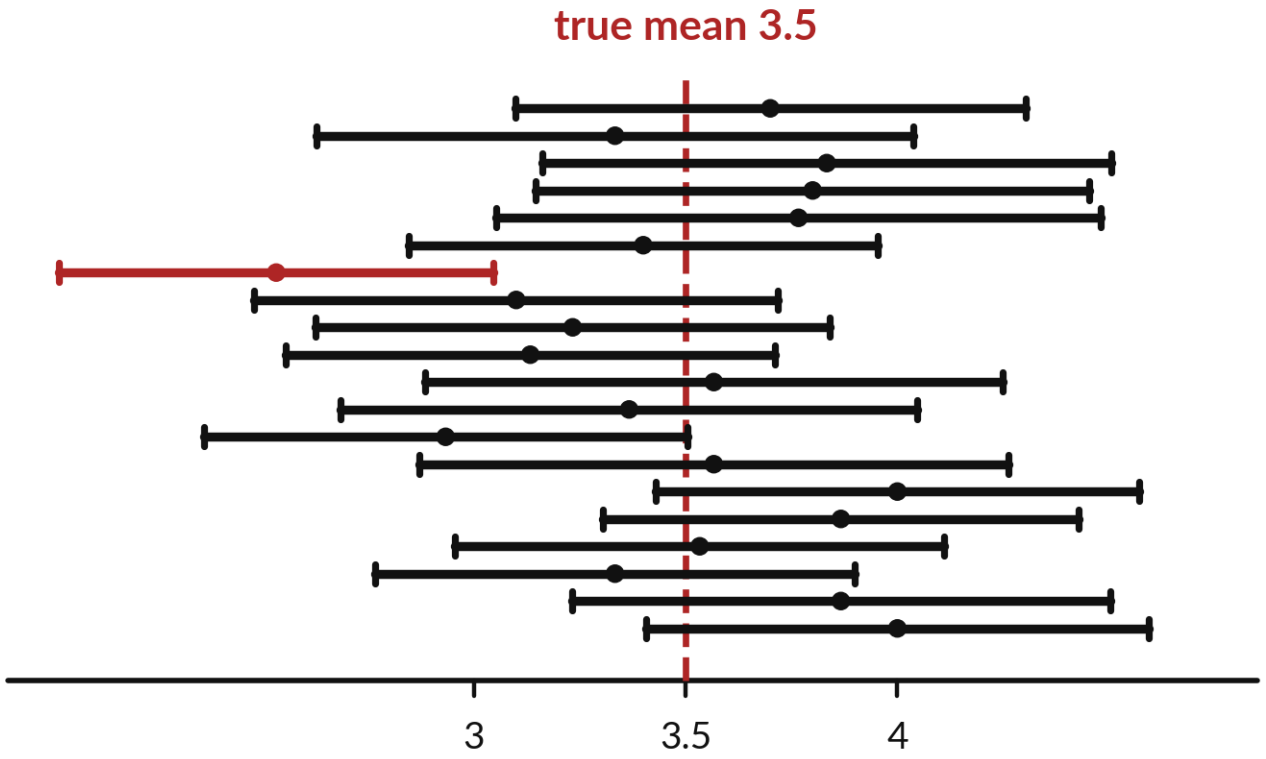
Standard error

$$\text{SE} = \frac{\sigma}{\sqrt{n}}$$

Symbol	What it means	In our data
x-bar	The mean of a sample of n values.	mean of 30 dice
sigma, n	Standard deviation of one value, and the sample size.	sigma = 1.7078 for a die
SE	Standard error: the standard deviation of the sample mean.	1.7078 / sqrt(30) = 0.3118

Four times the data halves the SE. Simulated 30-dice averages (seed 14): standard deviation 0.311.

A 95% confidence interval comes from a method that catches the true mean in about 95 of 100 samples



mean of 30 rolls, with its 95% interval

20 samples of 30 dice (simulation). 19 of the 20 intervals contain the true mean 3.5. The red one misses.

Interval = sample mean plus or minus 1.96 standard errors.

The 95% describes the method over many samples.

It is not the chance that this one interval holds the true value.

Confidence interval for a mean, worked on the 60 fixed rolls

Interval $\bar{x} \pm 1.96\,\mathrm{SE}, \qquad \mathrm{SE} = \frac{s}{\sqrt{n}}$

Symbol	What it means	In our data
x-bar, n	The mean of the 60 rolls from Session 13, and the sample size.	3.9667, 60
s	The sample standard deviation (divides by n - 1).	1.7847
SE	Standard error s over the square root of n.	0.2304
Interval	Mean plus or minus 1.96 SE.	3.515 to 4.418

1.96 is the normal approximation. With n = 60 the t value 2.00 gives 3.506 to 4.428, and 3.5 is still just outside.

An A/B test splits users at random into two groups and compares one rate

	A	B	C	D
	Group	**Users**	**Rated helpful**	**Rate**
1	A: current app	2,400	288	12.0%
2	B: new feature	2,400	336	14.0%

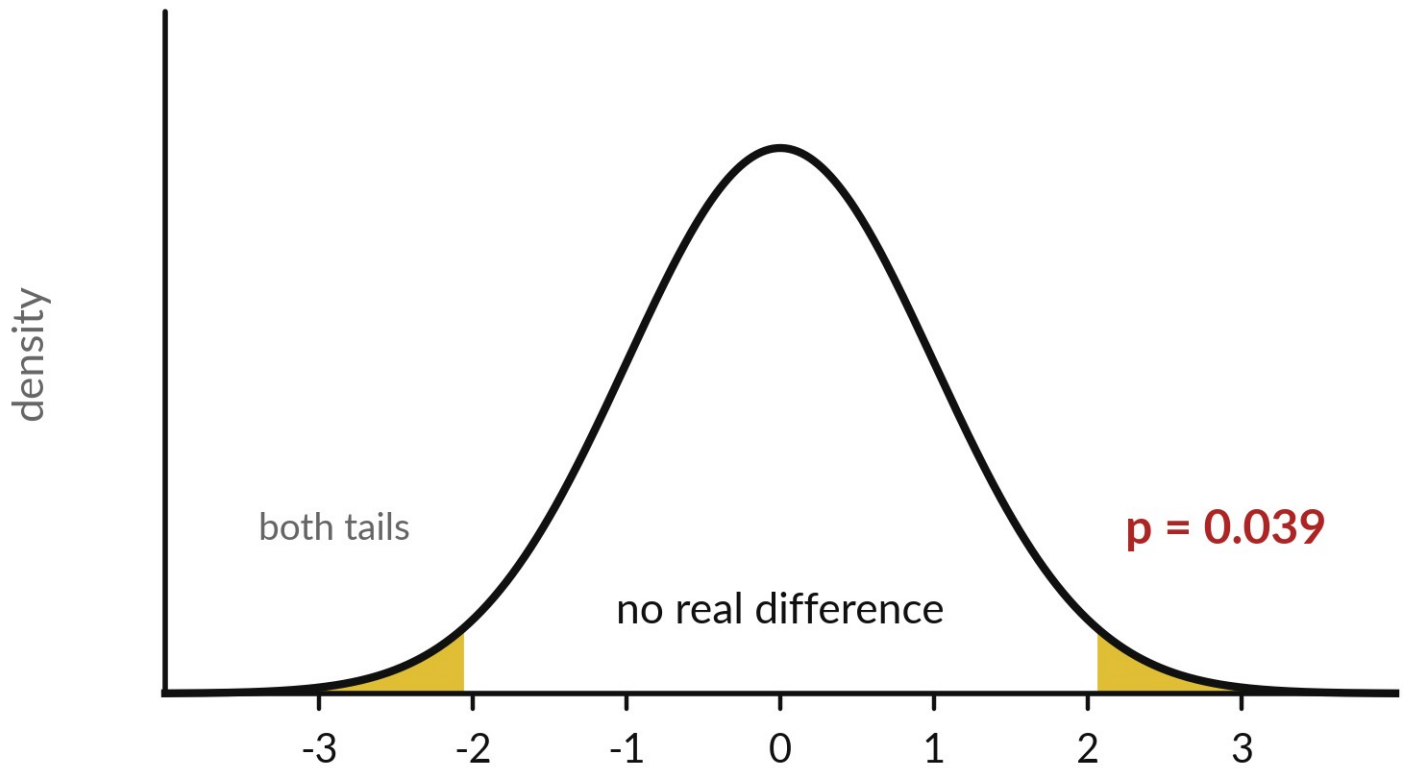
Illustrative counts: a new suggested follow-up feature in an AI chat app. Metric: share of chats rated helpful.

Random assignment makes the groups alike, apart from the feature.

B beats A by 2.0 percentage points.

Question: a real effect, or luck?

A hypothesis test asks how likely a gap this large would be if the feature did nothing



Null hypothesis: the true rates are equal. Our gap is z = 2.06 standard errors.

The gold tails hold gaps at least this large, in either direction.

Their total area is the p-value: 0.039.

Small p-value: this gap would be rare by luck alone.

The two-proportion z-test

Standard error under the null

$$\mathrm{SE} = \sqrt{\hat{p}_{\text{pool}}(1-\hat{p}_{\text{pool}})\left(\frac{1}{n_A}+\frac{1}{n_B}\right)}$$

z statistic

$$z = \frac{\hat{p}_B - \hat{p}_A}{\mathrm{SE}}$$

Symbol	What it means	In our data
p-hat_A, p-hat_B	The rate in each group: helpful chats k divided by users n.	288 / 2400 = 0.12, 336 / 2400 = 0.14
p-hat_pool	The pooled rate: (k_A + k_B) / (n_A + n_B), both groups together.	0.130
SE and z	The standard error of the gap, and the gap in standard errors.	0.00971 and 2.060

Two-sided p-value = 2 × (1 - area to the left of z) = 0.0394. Spreadsheet: 2*(1-NORM.DIST(z,0,1,TRUE)).

The A/B test step by step, with a 95% interval for the gap

Step	Calculation	Result
1 Rates	288 / 2400 and 336 / 2400	0.120 and 0.140
2 Pooled rate	(288 + 336) / (2400 + 2400)	0.130
3 Standard error	square root of 0.13 × 0.87 × (1/2400 + 1/2400)	0.00971
4 z statistic	0.020 / 0.00971	2.060
5 p-value	2 × (1 - area to the left of z)	0.0394
6 Interval for the gap	0.020 plus or minus 1.96 × 0.00970	0.0010 to 0.0390

Step 3 pools because the test assumes no difference. Step 6 does not assume that: SE = sqrt(p_A(1-p_A)/n_A + p_B(1-p_B)/n_B). The interval, 0.1 to 3.9 points, only just excludes 0.

A p-value is easy to misread

A p-value is

The probability of a gap at least this large, either way, if there were truly no difference

A p-value is not

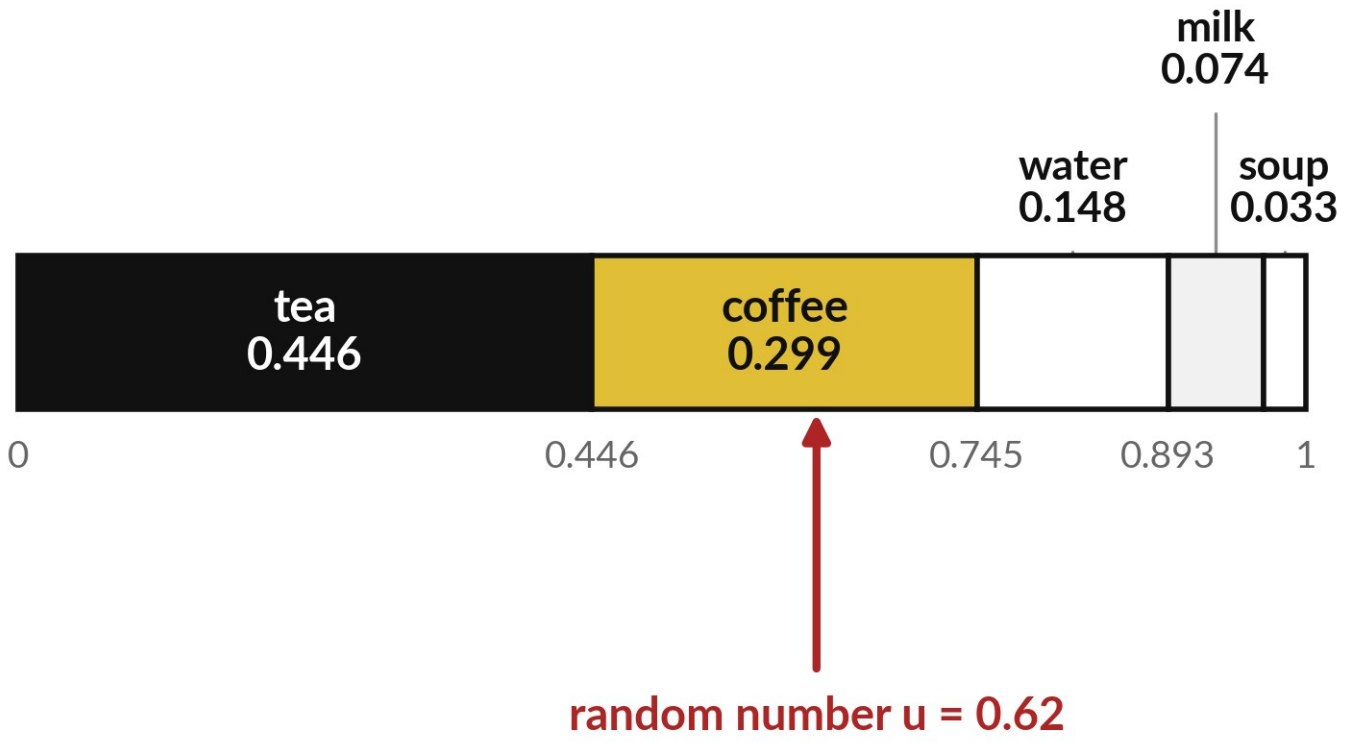
The probability that the feature does nothing

The size of the effect

Proof: test 20 useless features and about 1 passes at $p < 0.05$ by luck

Report the gap and its interval beside the p-value.

Sampling from a distribution: cut 0 to 1 into slices and see where a random number lands



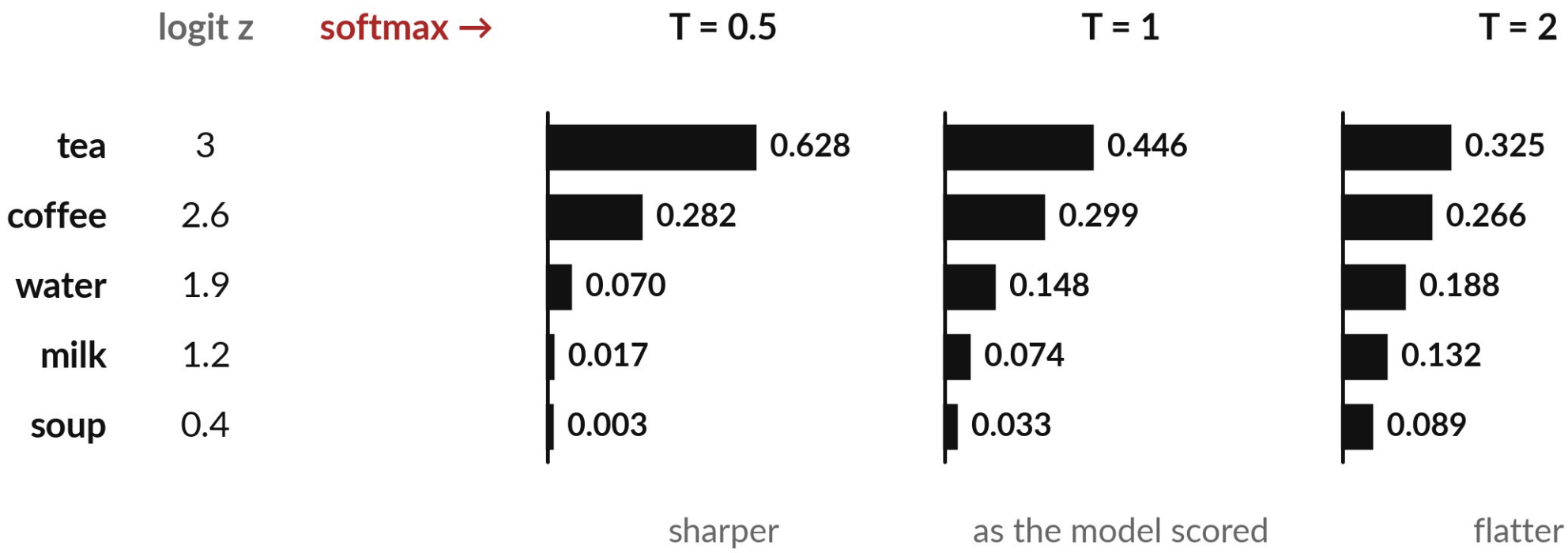
Each word owns a slice as wide as its probability. Draw a random number u between 0 and 1.

u = 0.62 lands in the coffee slice (0.446 to 0.745).

Draw again and u differs, so the word can differ.

Likely words are picked often, unlikely ones sometimes.

Temperature reshapes the probabilities: low is sharp, high is flat



Illustrative logits. Same five words, three temperatures. Each column of probabilities adds up to 1.

Softmax with temperature

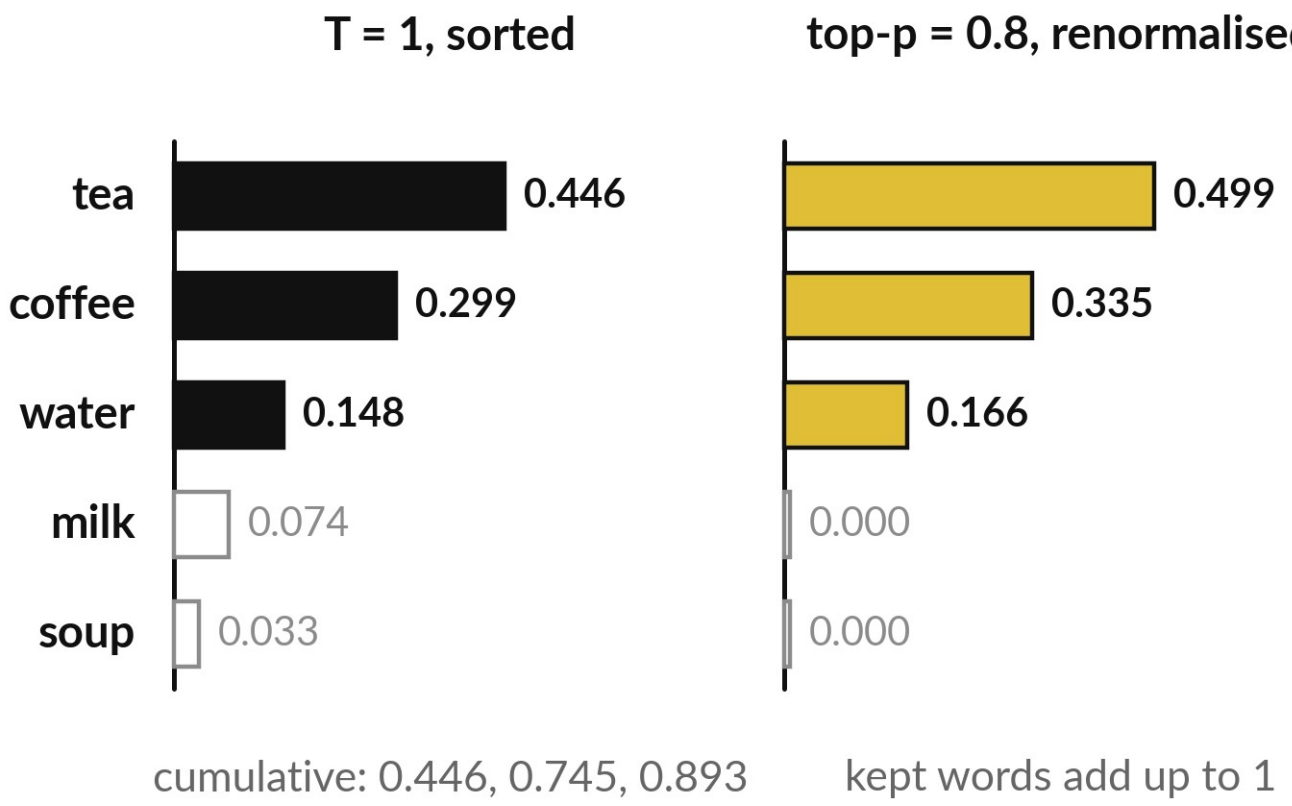
Softmax with temperature

$$p_i = \frac{e^{z_i/T}}{\sum_j e^{z_j/T}}$$

Symbol	What it means	In our data
z_i	The logit (raw score) of word i.	tea: z = 3.0
T	The temperature, above 0. T = 1 is the plain softmax.	T = 2
Worked	tea at T = 2: e to the 1.5 = 4.482. The five terms add up to 13.780.	4.482 / 13.780 = 0.3252
Limits	T near 0 always picks the top word (greedy). T very large makes every word equally likely.	T = 0.5 gives tea 0.628

Use the stable softmax of Session 12: subtract the largest z first. The result is the same.

Top-p (nucleus) sampling keeps only the smallest set of likely words, then renormalises



Top-p (nucleus) sampling with p = 0.8: add words from the most likely until the total reaches 0.8.

Tea, coffee and water reach 0.893. Milk and soup are cut (grey).

Divide the kept words by 0.893: 0.499, 0.335, 0.166.

Now milk and soup can never be chosen.

The same eight random numbers give different words at different temperatures

Draw	Random number u	T = 1	T = 0.5	T = 2
1	0.420	tea	tea	coffee
2	0.926	milk	water	soup
3	0.274	tea	tea	tea
4	0.060	tea	tea	tea
5	0.311	tea	tea	tea
6	0.718	coffee	coffee	water
7	0.781	water	coffee	milk
8	0.539	coffee	tea	coffee

Each word is the first slice whose running total reaches u. Illustrative probabilities, seed 19.

Bridge: sample the central limit theorem and the next word with NumPy

```
rng = np.random.default_rng(14)
means = rng.integers(1, 7, (20000, 30)).mean(axis=1)
means.mean(), means.std()
z = np.array([3.0, 2.6, 1.9, 1.2, 0.4])
p = np.exp(z / 0.5 - (z / 0.5).max())
p = p / p.sum()
rng.choice(5, p=p)
```

In plain words

Line 2: 20,000 samples of 30 dice. mean(axis=1) averages each row.

Line 3: the mean is near 3.5 and the standard deviation near the standard error 0.3118 (0.311 with seed 14).

Lines 4 to 6: softmax with temperature T = 0.5, stable version.

Line 7: draw one index from 0 to 4 with these probabilities.

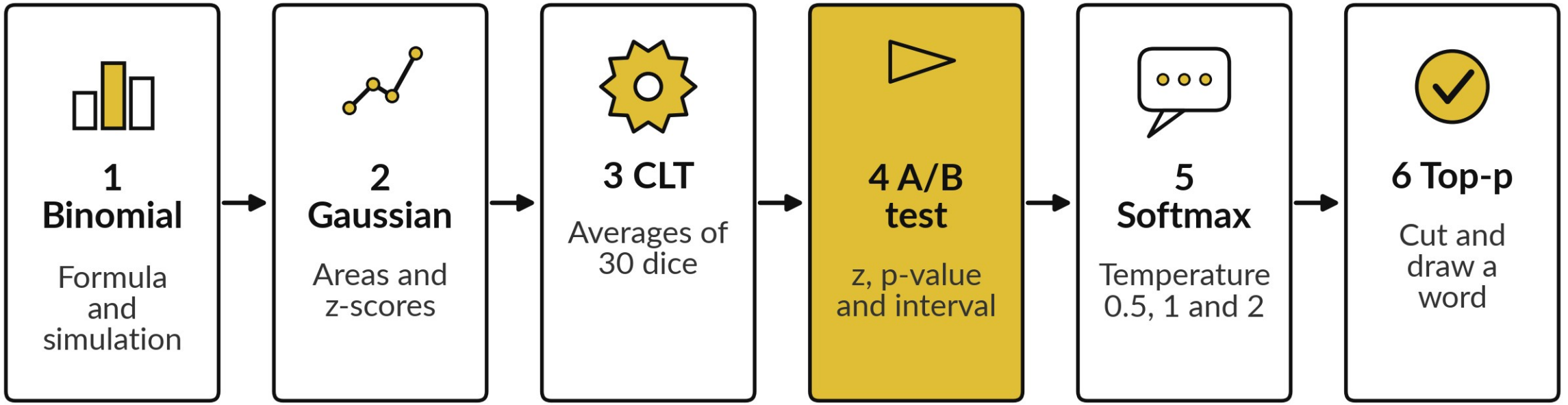
Spreadsheet: NORM.DIST, BINOM.DIST and fixed sample columns replace the random draws.



Part 2: the lab

Close the slides. Open the notebook or the workbook.

The lab has six stages, in Python and in the spreadsheet



Python notebook: Session 14. Workbook sheets: Binomial, Normal, Sampling, ABtest and Temperature. Predict, run, read, change one thing.

Check your numbers against these before you move on

What you compute	You should see
Binomial(10, 0.3): P(X = 3) by formula	0.2668
Normal(60, 10): P(X < 75) and P(50 < X < 70)	0.9332 and 0.6827
Standard error of the mean of 30 dice: formula, and simulated (seed 14)	0.3118; about 0.311
Confidence interval of the 60 fixed rolls	3.515 to 4.418
A/B test: pooled rate, z, two-sided p-value	0.13, 2.060, 0.0394
A/B test: 95% interval for the gap	0.0010 to 0.0390
P(tea) at T = 0.5, 1 and 2	0.6278, 0.4459, 0.3252
Top-p 0.8: renormalised P(tea)	0.4992

Simulated numbers use the seeds in the notebook. They match the slides only if the seed line runs first.

Errors you will meet, what they mean and what to do

	Where	What you see	What to do
✗	**Python**	Your sample numbers differ from the slides	Use Run all from the top. Every random step makes its own seeded generator.
✗	**Python**	p-value 0.0197 instead of 0.0394	You used one tail. Double the tail area for a two-sided test.
✗	**Spreadsheet**	BINOM.DIST gives P(X at most 3) = 0.6496	The last argument was TRUE. Use FALSE for exactly 3.
✗	**Spreadsheet**	NORM.DIST #VALUE!	It needs four arguments: x, mean, standard deviation and TRUE.
✗	**Both**	Probabilities after temperature do not add to 1	The sum must run over all five words, with the same T in every term.

Where this appears in AI: temperature and top-p are settings you can choose

Sampling settings in public APIs (names only)

temperature

Documented by OpenAI, Gemini, Anthropic

top_p (nucleus)

Documented by the same three APIs

top_k

Documented by some providers

Same question twice

Above T = 0, words are drawn at random

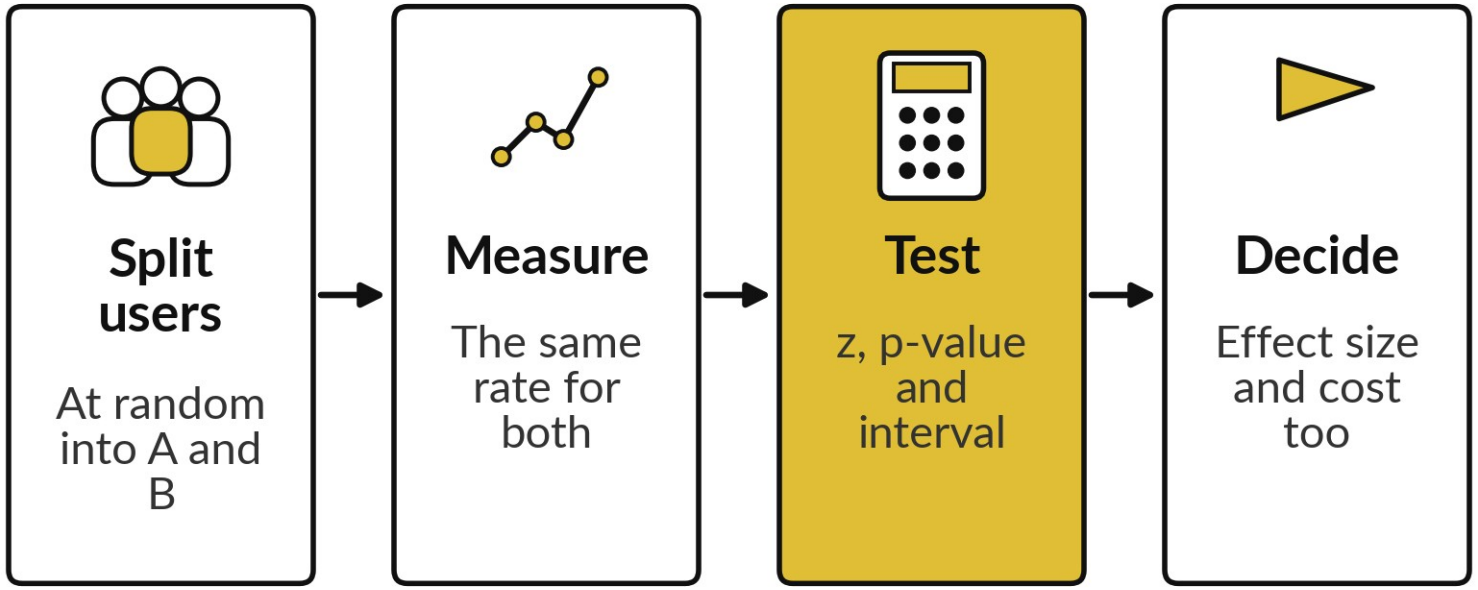
These settings control the sampling you did by hand today.

Low temperature: steadier answers. High: more varied.

Nucleus sampling: Holtzman and colleagues, 2020.

Only for models that expose these settings. Products may add steps we cannot see.

Where this appears in AI: an A/B test shows whether a new AI feature really is better



Online experiments compare two versions of a product on one measured rate.

Published practice: Kohavi, Tang and Xu describe it in their 2020 book on online controlled experiments.

The mathematics is the same as today. Which AI features a company tests is its own business.

Where this appears in AI: check an assistant with your numbers

8 minutes

Check an AI assistant

Ask an AI assistant to run the A/B test on our made-up counts.

Give it A: 288 of 2,400 and B: 336 of 2,400.

Ask for z, p and a 95% interval.

Compare with 2.06, 0.039 and 0.1 to 3.9 points.

Ask what p means. Look for misreadings.

You keep

A note: its numbers, your numbers, and any misreading you spotted.

Exit check: three questions

Question	Short answer
A fair coin is flipped 10 times. What are the mean and standard deviation of the number of heads?	n p = 5 and the square root of 10 × 0.5 × 0.5, which is 1.581.
A test gives p = 0.03. Does that mean a 3% chance the feature does nothing?	No. It is the chance of a gap this large if there is truly no difference.
What does lowering the temperature do to the next-word probabilities?	It makes them sharper: the top word gets more probability, and the answers repeat more.

Save the lab and keep your key numbers

What you keep from today

Keep your saved lab and these numbers: binomial $P(X = 3) = 0.2668$, normal $P(X < 75) = 0.9332$, A/B test $z = 2.06$ and $p = 0.039$, and the three softmax rows for $T = 0.5$, 1 and 2.

Our question today

Why can the same question get a different answer from a chat model, and how do we test whether a new AI feature is better?

Next session

Session 15: conditional probability and Bayes. We update beliefs with evidence and build a small spam filter by hand.

Take away

Saved lab
Notebook or workbook

Key numbers
See the check table

Next session
Bayes theorem

Bring
A question about a test result